JEE Mathematics: Quick Review with Grading and Practice Map - Area Under Curve (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (83 questions; a question that appears in two of these is counted once): High = 8 or more questions, Medium = 1 to 7 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Area Bounded by a Curve

Question 01

How to find area bounded by a curve and x-axis?

Model Answer

$$A = \int_a^b |y|\, dx$$

Area is always positive. If $f(x)$ changes sign at c, d in (a, b), split: $A = \int_a^c y\, dx - \int_c^d y\, dx + \int_d^b y\, dx$ $(= A_1 - A_2 + A_3$ with algebraic areas).

Ex: $y = \cos x$, $0 \le x \le 2\pi$: $A = \int_0^{2\pi} |\cos x|\, dx = 4$ (whereas $\int_0^{2\pi} \cos x\, dx = 0$).

Grade - High

Practice map, main concept: Sheet: J-I 1-4, 12-14, 19, 26, 36, 48-49, 51-53; J-II 9-10, 25, 29, 31, 34, 37; J-M 15, 20, 47; J-A 1, 4, 15; C-I 2-3, 7, 9-11, 13; C-II 8, 12, 16, 21, 28 | Question Bank: 154, 173, 176, 180

Also used in: Sheet: J-I 23, 37; J-II 4, 20-22; J-M 6, 29, 35; J-A 14; C-II 9, 14, 29 | Question Bank: 156

Question 02

How to find the area of an unbounded region (a region that extends to infinity)?

Model Answer

$$A = \int_a^\infty |f(x)|\, dx = \lim_{b\to\infty} \int_a^b |f(x)|\, dx$$

The area is finite only if the limit exists (e.g. $f \to 0$ fast enough, like e^{-x} or $\frac{1}{x^2}$); near a vertical asymptote use a limit at that end.

Ex: $y = \frac{1}{1+x^2}$ and the x-axis: $A = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 7, 36; C-I 10; C-II 28

Area Bounded between two curves

Question 03

How to find area bounded between two curves?

Model Answer

$$A = \int_a^b |f(x) - g(x)|\, dx$$

Sketch both curves, find the points of intersection, then integrate (upper − lower); if the curves cross at $x = c$, split the integral at c.

Ex: $y = 2^x$, $y = 2x - x^2$, $x = 0$, $x = 2$: $A = \int_0^2 (2^x - 2x + x^2)\, dx = \frac{3}{\ln 2} - \frac{4}{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 5-11, 15-17, 20-22, 25, 28, 30-35, 50; J-II 1-3, 5, 7-8, 13-15, 23, 33, 35-36; J-M 1-2, 9-11, 13, 16, 19, 23, 26-28, 30-31, 34, 36, 38-39, 41-42, 44-46; J-A 2, 5, 8-9, 11; C-I 1, 4-5, 12; C-II 1-4, 6-7, 10, 19-20, 30 | Question Bank: 150-151, 153, 159, 161, 163-166, 168, 170, 172, 174, 178-179, 183, 186, 188, 190-191

Also used in: Sheet: J-I 18, 38, 47; J-II 16, 18-19; J-M 3-5, 8, 12, 14, 17-18, 21-22, 24-25, 32-33, 37, 40, 43; J-A 7, 10, 13; C-II 5, 11, 13, 15, 17 | Question Bank: 149, 152, 158, 160, 162, 167, 171, 175, 177, 181-182, 184, 187, 189

Question 04

Can you plot graph of a polynomial function (linear, quadratic, cubic or higher degree)?

Model Answer

Mark the roots (sign changes at odd-multiplicity roots, touching at even), the y-intercept, the end behaviour (sign of leading coefficient) and the turning points from $f'(x) = 0$.

Quadratic $y = ax^2 + bx + c$: vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$, opens up if $a > 0$.

Ex: $y = x^3 - 3x$: roots $0, \pm\sqrt{3}$, max 2 at $x = -1$, min -2 at $x = 1$; area with the x-axis $= 2\int_0^{\sqrt{3}} (3x - x^3)\, dx = \frac{9}{2}$.

Grade - High

Practice map, main concept: none

Also used in: Sheet: J-I 3, 6-7, 13, 16, 23, 37-38, 47; J-II 5, 15, 22, 25, 37; J-M 15, 19, 23, 35, 39; J-A 7, 9-10, 12-15; C-I 7, 12; C-II 1, 29 | Question Bank: 160, 164, 172, 174, 176, 178, 182, 184, 188, 190

Question 05

Can you plot graph of a standard second degree curve (Circle, Parabola, Ellipse or Hyperbola)?

Model Answer

$x^2 + y^2 = r^2$: centre O, radius r, area πr^2. $y^2 = 4ax$: vertex O, opens right ($a > 0$); $x^2 = 4ay$ opens up.

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: vertices $(\pm a, 0), (0, \pm b)$, area πab. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$: vertices $(\pm a, 0)$, asymptotes $y = \pm\frac{b}{a}x$.

Ex: area of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is $\pi \cdot 3 \cdot 2 = 6\pi$.

Grade - High

Practice map, main concept: Sheet: J-II 11

Also used in: Sheet: J-I 1, 4-5, 17, 24, 27, 32, 35, 44-47; J-II 1, 5, 13-14; J-M 4, 8, 11, 17-18, 21, 27, 29, 34, 38-40, 44-45; J-A 5, 8, 10, 16; C-I 1, 6, 8; C-II 2-4, 18, 31 | Question Bank: 149, 152-153, 161, 163-164, 168-169, 174, 177, 179, 183, 187, 191

Question 06

Can you plot graph of a shifted second degree curve (Parabola, Ellipse or Hyperbola)?

Model Answer

Complete the squares: $(y-k)^2 = 4a(x-h)$ is $y^2 = 4ax$ with vertex moved to (h,k); $\frac{(x-h)^2}{a^2} \pm \frac{(y-k)^2}{b^2} = 1$ has centre (h,k).

Ex: $y^2 - 2y + 4x + 5 = 0 \Leftrightarrow (y-1)^2 = -4(x+1)$: vertex $(-1,1)$, opens left.

Grade - High

Practice map, main concept: none

Also used in: Sheet: J-I 8, 11, 17, 21, 30, 33-34, 44, 46; J-II 2, 6, 11, 13-14; J-M 1-3, 6-7, 9-10, 12, 18, 22, 24, 26-28, 30-33, 36, 38, 46; J-A 16; C-II 30-31 | Question Bank: 150-151, 153, 155, 162, 170-171, 181, 186-187, 189, 191

Question 07

Can you plot graph of six trigonometric functions?

Model Answer

$\sin x, \cos x$: period 2π, range $[-1,1]$. $\tan x, \cot x$: period π, range $\mathbb{R}$. $\sec x, \csc x$: period 2π, range $(-\infty,-1] \cup [1,\infty)$.

Asymptotes: $\tan x, \sec x$ at $x = (2n+1)\frac{\pi}{2}$; $\cot x, \csc x$ at $x = n\pi$.

Ex: $y = \sin x$, $y = \cos x$ and the y-axis (1st quadrant): $A = \int_0^{\pi/4} (\cos x - \sin x)\,dx = \sqrt{2} - 1$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 2, 9, 12, 18, 22, 31, 36, 48-49; J-II 4, 8; J-M 5, 20, 47; J-A 2; C-I 2; C-II 10, 16-17 | Question Bank: 173, 180

Question 08

Can you plot graph of six ITFs?

Model Answer

$\sin^{-1} x$: $[-1,1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\cos^{-1} x$: $[-1,1] \to [0,\pi]$; $\tan^{-1} x$: $\mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$; $\cot^{-1} x$: $\mathbb{R} \to (0,\pi)$.

$\sec^{-1} x$: $|x| \geq 1 \to [0,\pi] - \{\frac{\pi}{2}\}$; $\csc^{-1} x$: $|x| \geq 1 \to [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$. Each graph is the mirror image of the restricted trig graph in $y = x$.

Ex: $y = \sin^{-1} x$, $y = \cos^{-1} x$ and the x-axis: integrate in y: $A = \int_0^{\pi/4} (\cos y - \sin y)\,dy = \sqrt{2} - 1$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 6

Question 09

Can you plot graph of e^x, $\ln x$, $\{x\}$, $[x]$, $|x|$?

Model Answer

$e^x > 0$ through $(0,1)$, asymptote $y = 0$; $\ln x$ $(x > 0)$ through $(1,0)$, asymptote $x = 0$ (mirror images in $y = x$).

$[x]$: steps, $[x] = k$ on $[k, k+1)$; $\{x\} = x - [x]$: saw-tooth in $[0,1)$, period 1; $|x|$: V-shape with vertex O.

Ex: $y = \ln x$, x-axis, $x = \frac{1}{e}$, $x = e$: $A = -\int_{1/e}^{1} \ln x\,dx + \int_1^e \ln x\,dx = 2 - \frac{2}{e}$.

Grade - High

Practice map, main concept: none

Also used in: Sheet: J-I 5, 8, 10, 15, 20, 25-26, 28, 32, 39-43; J-II 3, 7, 18, 33; J-M 8, 13-16, 23, 41, 43; J-A 2, 5-6, 11; C-I 2-3, 5; C-II 28, 31 | Question Bank: 154, 156, 158-160, 162, 165, 175-176, 179, 188

Question 10

Revise transformation of graphs taught in function.

Model Answer

$f(x+a)$: shift left by a; $f(x)+a$: shift up; $f(-x)$: mirror in y-axis; $|f(x)|$: reflect the part below the x-axis; $f(|x|)$: keep $x \geq 0$ part and mirror it.

Ex: $|x|+|y| \leq a$ is a square with vertices $(\pm a, 0), (0, \pm a)$: area $2a^2$.

Grade - **High**

Practice map, main concept: none

Also used in: Sheet: J-I 3, 5, 10, 19-20, 25-26, 28, 39-41, 43; J-II 7, 13; J-M 5, 13-14, 16, 21, 25, 41; J-A 6, 11, 13; C-I 6, 8, 12; C-II 6, 29, 31 | Question Bank: 159, 163, 165

Question 11

Revise plotting of graph taught in Monotonicity.

Model Answer

Find domain, symmetry, intercepts, sign of f' (increasing / decreasing), extrema and asymptotes; for $y^2 = g(x)$ the curve is symmetric about the x-axis and exists only where $g(x) \geq 0$.

Ex: loop of $ay^2 = x^2(a-x)$ $(0 \leq x \leq a)$: $A = 2\int_0^a x\sqrt{1-\frac{x}{a}}\,dx = \frac{8a^2}{15}$.

Grade - **Medium**

Practice map, main concept: none

Also used in: Sheet: J-I 14, 19, 50; J-II 3, 9-10, 12, 29, 36-37; J-M 42; J-A 14; C-I 9-10; C-II 12, 28 | Question Bank: 161, 166

Question 12

How to write expression of integral to find area when integrating wrt x-axis?

Model Answer

$$A = \int_{x_1}^{x_2} (y_{\text{upper}} - y_{\text{lower}})\,dx$$

Vertical strip of width dx: write both boundaries as $y = f(x)$; if the upper or lower boundary changes (a new curve takes over), split at that x.

Ex: $y = x^2$ and $y = 2x$: $A = \int_0^2 (2x - x^2)\,dx = \frac{4}{3}$.

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 1

Integrating wrt y-axis

Question 13

How to write expression of integral to find area when integrating wrt y-axis?

Model Answer

$$A = \int_c^d (x_{\text{right}} - x_{\text{left}})\,dy$$

Horizontal strip of width dy: write each boundary as $x = f(y)$. Use it when the curves are $x = f(y)$ (e.g. $y^2 = 4ax$) or when vertical strips need splitting.

Ex: $x = 2y - y^2$ and the y-axis: $A = \int_0^2 (2y - y^2)\,dy = \frac{4}{3}$.

Grade - **High**

Practice map, main concept: none

Also used in: Sheet: J-I 17, 27, 30, 32-35, 45, 50; J-II 17, 33; J-M 1, 9, 18, 22, 27-28, 32, 36, 38, 40, 45; C-II 5, 19, 30 | Question Bank: 150, 161, 168, 171, 175, 177, 181, 186-187, 191

Important Results

Question 14

Write result for area bounded between two standard parabolas.

Model Answer

$$y^2 = 4ax,\ x^2 = 4by\ (a, b > 0): \quad A = \frac{16ab}{3}$$

Ex: $y = \sqrt{x}$ and $x = \sqrt{y}$, i.e. $y^2 = x$, $x^2 = y$ $(a = b = \frac{1}{4})$: $A = \frac{16}{3} \cdot \frac{1}{16} = \frac{1}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 27; C-I 6, 8; C-II 29 | Question Bank: 155

Also used in: Sheet: J-I 29

Question 15

Write result for area bounded between standard parabola and line $y = mx$.

Model Answer

$$y^2 = 4ax,\ y = mx: \quad A = \frac{8a^2}{3m^3}$$

Ex: $y^2 = 4x$, $y = x$: $A = \int_0^4 (2\sqrt{x} - x)\, dx = \frac{8}{3}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 32

Question 16

How to find area between a parabola (whose axis is horizontal/vertical) and a line?

Model Answer

$$y = ax^2 + bx + c \text{ and a line meeting at } x = \alpha, \beta: \quad A = \frac{|a|(\beta - \alpha)^3}{6}$$

Same with x and y swapped for a horizontal axis ($x = ay^2 + by + c$, roots in y). Also: a parabolic segment cut off by a chord $= \frac{2}{3}$ of the enclosing rectangle (e.g. $y^2 = 4ax$ and the double ordinate $x = a$).

Ex: $y^2 + 4x = 4$, $y - 2x = 2$: meet at $y = -4, 2$; $A = \frac{1}{4} \cdot \frac{6^3}{6} = 9$.

Grade - High

Practice map, main concept: Sheet: J-M 37 | Question Bank: 185

Also used in: Sheet: J-I 6, 16, 33, 45; J-II 15-17, 19, 23, 35; J-M 1, 9-10, 28, 30, 38, 40, 46; C-II 2, 4-5, 15, 29-30 | Question Bank: 151

Area represented by Inequalities

Question 17

How to find area of a region represented by inequalities?

Model Answer

Draw each boundary curve (replace $\leq$ by $=$), test a point to shade the correct side, take the common region, find its corner points, then integrate.

Ex: $\{(x, y) : 5x^2 \leq y \leq 2x^2 + 9\}$: curves meet at $x = \pm\sqrt{3}$; $A = \int_{-\sqrt{3}}^{\sqrt{3}} (9 - 3x^2)\, dx = 12\sqrt{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 40-47; J-M 3-6, 8, 12, 14, 17-18, 21-22, 24-25, 29, 32-33, 35, 40, 43; J-A 6, 10, 12-13, 16; C-II 31 | Question Bank: 149, 152, 156-158, 160, 162, 167, 169, 171, 175, 177, 181-182, 184, 187, 189

Also used in: Sheet: J-M 2; J-A 7

Shifting of Origin

Question 18

What is the effect on area on shifting of origin?

Model Answer

None: area is invariant when the axes are shifted ($X = x - h$, $Y = y - k$). So reduce shifted conics to standard form and use the standard results.

Ex: $y^2 - 2y + 4x + 5 = 0$, $x^2 + 2x - y + 2 = 0$ become $Y^2 = -4X$, $X^2 = Y$; $A = \frac{16 \cdot 1 \cdot \frac{1}{4}}{3} = \frac{4}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 39

Also used in: Sheet: J-I 21; J-II 6; J-A 6 | Question Bank: 155, 158

Question 19

What are the things to keep in mind while using shifting of origin in finding area?

Model Answer

Shift every boundary with the same h, k: other curves, lines, limits, and the axes themselves (x-axis $y = 0$ becomes $Y = -k$). Only the shape is kept, not the equations.

Ex: $(y - 2)^2 = x - 1$, its tangent at $(2, 3)$ and the x-axis: $Y^2 = X$, tangent $X = 2Y - 1$, x-axis $Y = -2$; $A = \int_{-2}^{1} (Y - 1)^2\, dY = 9$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 6

Area bounded by inverse of a function

Question 20

What is the effect on area if we take image of whole region wrt a line?

Model Answer

None: reflection keeps lengths, so the image region has the same area. Hence area under f^{-1} = area of the reflected region next to f.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 1, 6

Question 21

How to find area of a region by taking its image wrt a line?

Model Answer

For $g = f^{-1}$: reflect in $y = x$; the region under g between $x = p$, $x = q$ becomes the region between $y = f(x)$ and the y-axis from $y = p$ to $y = q$; find $f(\alpha) = p$, $f(\beta) = q$ and integrate f w.r.t. x.

$$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = bf(b) - af(a)$$

Ex: $f = x^3 + 3x + 2$, $g = f^{-1}$, area of g, x-axis, $x = -2$, $x = 6$: $f(-1) = -2$, $f(0) = 2$, $f(1) = 6$; $A = \int_{-1}^{0}(f + 2)\,dx + \int_0^1(6 - f)\,dx = \frac{9}{2}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 22

What are the equations of common lines about which image of region can be taken?

Model Answer

$y = x$: $(x, y) \to (y, x)$ (inverse function); $y = -x$: $(x, y) \to (-y, -x)$; $x = a$: $(x, y) \to (2a - x, y)$; $y = b$: $(x, y) \to (x, 2b - y)$.

f and f^{-1} are mirror images in $y = x$: area between them = 2× area between f and $y = x$.

Ex: $y = x^2$ $(x \geq 0)$ and $y = \sqrt{x}$: $A = 2\int_0^1(x - x^2)\,dx = \frac{1}{3}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Variable Area

Question 23

What is the condition on c for area bounded by $y = f(x)$, $x = a$, $x = b$ and $y = f(c)$ to be minimum if f is monotonic?

Model Answer

$$c = \frac{a + b}{2}$$

$A(c) = \int_a^c |f(c) - f|\,dx + \int_c^b |f - f(c)|\,dx$, $A'(c) = f'(c)\,[2c - (a + b)] = 0$; $A' < 0$ before and > 0 after the midpoint.

Ex: $f = \frac{x^3}{3} - x^2 + a$ (monotonic on $(0, 2)$), $x = 0$, $x = 2$ and the x-axis, area least $\Rightarrow f(1) = 0 \Rightarrow a = \frac{2}{3}$.

Grade - Low

Practice map, main concept: Sheet: J-II 12; C-II 18

Also used in: none

Question 24

How to find a parameter when the area of a region (or a relation between areas) is given?

Model Answer

Find the intersection points in terms of the parameter, write the area as a function of it, equate to the given value and solve; reject roots for which the sketch (region, sign, order of limits) changes.

Ex: area between $y = x^2$ and $y = ax$ $(a > 0)$ is $\frac{9}{2}$: $\int_0^a (ax - x^2)\,dx = \frac{a^3}{6} = \frac{9}{2} \Rightarrow a = 3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 29; J-II 16-18, 22; J-M 7; C-II 5, 11, 13, 17

Also used in: Sheet: J-M 10, 43; J-A 9, 15 | Question Bank: 156, 167

Question 25

How to find the line (or curve) that divides a region into two parts of given ratio, e.g. bisects its area?

Model Answer

Find the total area A; write the area of one part in terms of the parameter of the dividing line and set it equal to $\frac{m}{m+n}A$ (bisect: $\frac{A}{2}$). Choose the strip for which the part is a single integral.

Ex: $y = c$ bisects the region between $y = x^2$ and $y = 1$ (area $\frac{4}{3}$): $\frac{4}{3}c^{3/2} = \frac{2}{3} \Rightarrow c = \frac{1}{2^{2/3}}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18, 37-38; J-II 4, 6, 21; J-A 7, 14

Also used in: Sheet: J-II 5; C-II 17

Question 26

How to find the greatest or least value of an area that depends on a parameter?

Model Answer

Write $A(k)$ by integration, then use $A'(k) = 0$ with the sign of A' (or a perfect-square / AM-GM argument). For a parabola and a line through a fixed point, $A = \frac{|a|(\beta-\alpha)^3}{6}$ is least when $(\beta - \alpha)^2$ is least.

Ex: $y = x^2 + 2x - 3$ and $y = kx + 1$: $A = \frac{[(k-2)^2+16]^{3/2}}{6}$, least at $k = 2$, $A_{\min} = \frac{32}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23; J-II 19-20; C-II 9, 14-15

Also used in: Sheet: J-II 17; J-M 37-38 | Question Bank: 185-186

Average Value of a Function

Question 27

How to find average value of a function over an interval?

Model Answer

$$f_{\text{av}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

It can be positive, negative or zero; on $(0, \infty)$: $f_{\text{av}} = \lim_{b\to\infty} \frac{1}{b}\int_0^b f\,dx$ if the limit exists.

Ex: average of $\sin x$ on $[0, \pi] = \frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{2}{\pi}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 28
How to find RMS value of a function over an interval?

Model Answer

$$f_{\text{rms}} = \left(\frac{1}{b-a} \int_a^b f^2(x)\, dx \right)^{1/2}$$

Ex: RMS of $\sin x$ on $[0, 2\pi] = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 x\, dx} = \frac{1}{\sqrt{2}}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Area when curve is given in parametric form

Question 29
How to find area of a curve which is given parametrically?

Model Answer

$$x = x(t),\ y = y(t): \quad A = \left| \int_{t_1}^{t_2} y(t)\, x'(t)\, dt \right|$$

Change the x-limits into t-limits; for a closed curve integrate over one full round (or use symmetry).
Ex: $x = 2\cos\theta,\ y = 2\sin\theta$: $A = \left| \int_0^{2\pi} 2\sin\theta(-2\sin\theta)\, d\theta \right| = 4\pi$.

Grade - Low
Practice map, main concept: Sheet: J-I 24
Also used in: none